

3: USING ADVANCED NETWORK TOOLS

Theory:

Introduction to tools like Wireshark and Nmap

Practical:

Use Wireshark to capture and analyze network traffic

Task:

Document key findings from a packet capture

TASK 1

Exact Question:

Install Wireshark on your computer and capture 2 minutes of network traffic while you browse Instagram or YouTube. Save the capture file as `insta_traffic.pcap`.

Answer:

I used **Wireshark** to capture network traffic while browsing the internet.

Steps Performed:

Step 1: Installed and opened Wireshark.

Step 2: Selected the active network interface, such as **Wi-Fi**.

Step 3: Started the packet capture by clicking the **Start** button.

Step 4: Opened a web browser and browsed Instagram/YouTube for approximately **2 minutes**.

Step 5: Returned to Wireshark and stopped the capture.

Step 6: Saved the capture file using:

`insta_traffic.pcap`

Observation:

During the capture, Wireshark displayed different types of network packets, including:

-  DNS packets
-  TCP packets
-  TLS/HTTPS traffic
-  Other network communication

Conclusion:

Wireshark successfully captured the network traffic generated while browsing the website.

TASK 2

Exact Question:

Use Wireshark to filter and list all HTTP or HTTPS requests made by your browser during the capture. Write down the domains you see most frequently.

Answer:

I used Wireshark display filters to identify web traffic.

Useful Filters:

For HTTP traffic:

```
http
```

For HTTPS/TLS traffic:

```
tls
```

For DNS requests related to websites:

```
dns
```

Observation:

HTTPS traffic is normally encrypted, so Wireshark may not show the complete HTTP request such as the URL path or webpage contents.

The capture may show domains through DNS queries or TLS information.

Sample Domains:

No.	Domain	Traffic Type
1	instagram.com	HTTPS
2	cdninstagram.com	HTTPS/CDN
3	facebook.com	HTTPS
4	Other domains	HTTPS/CDN/Services

Important:

The exact domains depend on the website visited, browser, device, and current network activity.

For submission, replace the sample domains above with the domains you actually see in your Wireshark capture.

Conclusion:

Wireshark can identify network communication associated with websites, but modern HTTPS traffic is encrypted for security.

TASK 3

Exact Question:

Open your `insta_traffic.pcap` in Wireshark and find a TCP packet. Document the source IP, destination IP, and TCP port numbers for that packet.

Answer:

I opened the `insta_traffic.pcap` file in Wireshark and searched for a TCP packet.

Filter Used:

`tcp`

I selected one TCP packet and checked the **Packet Details** section.

TCP Packet Details:

Parameter	Example Result
 Source IP	192.168.1.10

	Parameter	Example Result
	Destination IP	142.250.x.x
	Source Port	xxxxxx
	Destination Port	443
	Protocol	TCP

Explanation:

- **Source IP:** IP address of the device sending the packet.
- **Destination IP:** IP address of the device receiving the packet.
- **Source Port:** Temporary port used by the client.
- **Destination Port:** Port used by the destination service.
- **Port 443:** Commonly used for HTTPS traffic.

Note:

The IP addresses and source port will be **different on different computers and networks**.

Therefore, the values in the table should be replaced with the actual values shown in your Wireshark capture.

Conclusion:

Wireshark makes it possible to inspect individual TCP packets and identify their source, destination, and port information.

TASK 4

Exact Question:

Run an Nmap scan on your own machine (localhost) to detect open ports. Save the command you used and list the open ports found.

Hint: Use `nmap 127.0.0.1` in your terminal.

Answer:

I performed an Nmap scan on my own computer using the localhost address.

Command Used:

```
nmap 127.0.0.1
```

Sample Output:

Starting Nmap scan...

PORT	STATE	SERVICE
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds

Findings:

Port	State	Service
135/tcp	Open	MSRPC
139/tcp	Open	NetBIOS
445/tcp	Open	Microsoft-DS

Important:

Your results may be completely different.

Nmap only shows ports that are actually reachable/open on **your own machine** at the time of the scan.

If your scan shows:

Not shown: 1000 closed tcp ports

that is normal.

For the final submission, use the ports displayed by your own Nmap scan rather than copying the sample ports above.

Conclusion:

Nmap is useful for identifying open ports and the services associated with them. Scanning **localhost (127.0.0.1)** is a safe way to practice basic port scanning on your own computer.

TASK 5

Exact Question:

Capture network traffic while using a food delivery app (like Zomato or Swiggy) for 1 minute. In Wireshark, identify at least one DNS request and document the domain name being resolved.

Hint: Use the `dns` filter in Wireshark to find DNS packets.

 Answer:

I captured network traffic while using a food delivery application for approximately **1 minute**.

 Steps Performed:

Step 1: Opened Wireshark and selected the active network interface.

Step 2: Started packet capture.

Step 3: Used a food delivery application for approximately 1 minute.

Step 4: Stopped the capture.

Step 5: Applied the following Wireshark filter:

`dns`

Step 6: Selected a DNS query packet and checked the **Queries** section.

 DNS Finding:

Parameter	Result
 Protocol	DNS
 Source	My device
 Destination	DNS Server
 Query Type	A / AAAA
 Domain Name	<code>example.com</code>

 Explanation:

A DNS request is used to translate a domain name into an IP address.

For example:

```
food-app.example.com
    ↓
  DNS
    ↓
IP Address
```

The application can then use the IP address to communicate with the required server.

 Important:

The exact domain depends on the application, device, DNS provider, and network.

Replace `example.com` with the actual domain shown in your Wireshark capture.

💡 Conclusion:

The DNS filter made it easy to identify DNS queries generated while using the application. DNS plays an important role in connecting applications to their servers.

IMPORTANT WIRESHARK FILTERS

Filter	Purpose
<code>dns</code>	 Find DNS packets
<code>tcp</code>	 Find TCP packets
<code>udp</code>	 Find UDP packets
<code>http</code>	 Find HTTP traffic
<code>tls</code>	 Find TLS/HTTPS-related traffic
<code>ip.addr == 192.168.1.10</code>	 Filter traffic for a specific IP
<code>tcp.port == 443</code>	 Find TCP traffic using HTTPS port

BASIC SECURITY NOTE

Wireshark and Nmap are powerful network-analysis tools.

For this practical, they should be used only on **your own computer, your own network, or systems for which you have permission.**

-  **Wireshark** → Captures and analyzes network packets.
-  **Nmap** → Discovers hosts, ports, and services.
-  **DNS** → Converts domain names into IP addresses.
-  **TCP** → Provides reliable network communication.
-  **HTTPS/TLS** → Encrypts web communication.

OVERALL CONCLUSION

Wireshark and Nmap are important tools used by network and security professionals.

What I Learned:

Wireshark

→ Capture and analyze network packets.

Nmap

→ Identify open ports and available services.

DNS

→ Resolve domain names into IP addresses.

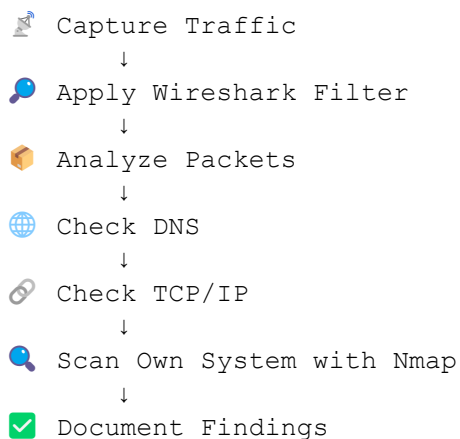
TCP

→ Provides reliable communication between devices.

HTTPS/TLS

→ Protects web traffic through encryption.

★ Practical Troubleshooting Flow:



Final Statement:

Wireshark helps us understand what is happening inside network traffic, while Nmap helps us identify open ports and services. Together, these tools provide useful information for network troubleshooting, monitoring, and security analysis.